Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/01/2007
PI Name:	Fedorov, Andrei G Ph.D.		
Project Title:	MEMS-ENABLED PROCESSING OF LIQUID FUELS FOR DISTRIBUTED POWER GENERATION USING FUEL CELLS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30332	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2003	**End Date:**	12/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2007 per PI (11/07) NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Degertekin, Levent (Georgia Institute of Technology)		
Grant/Contract No.:	NNC04GA12G		
Performance Goal No.:			
Performance Goal Text:			

	Fuel cell based portable power sources offer opportunities for high efficiency and ultra-high density distributed power generation, which is essential for enhanced capabilities and survivability of the future long duration space missions as well as for extra vehicular activities (EVA) of astronauts. A key element of the fuel cell power sources is the fuel processor, which should have a capability to produce hydrogen from a variety of hydrocarbon fuels stored in a liquid phase for highest power density. To this end, this project focuses on the novel portable MEMS hydrogen generator with improved functionality and low cost achieved through innovative system integration approach and by exploiting advantages pertinent to fluid mechanics, heat and mass transfer, and catalysis on micro/nano scale. The proposed microreactor has multifuel processing capability achieved by using unique piezoelectrically-driven fuel atomization technology, it operates in autothermal regime that does not require external heating, it exhibits high fuel conversion and selectivity to hydrogen, and it features low “skin” temperature which is essential from thermal management prospective. To establish the fundamental bases for optimal design and operation of the proposed catalytic microreactor, a ground-based three-year experimental study of the microreactor and its key components is undertaken. The key objectives include (1) design, fabrication, and testing of the autothermal MEMS catalytic reactor integrated with hydrogen separating membrane, (2) design, fabrication, and experimental characterization of the novel micromachined fuel ejector array for energy-efficient atomization and mixing of liquid fuels, and (3) integration and experimental characterization of an integrated microreformer that combines a microreactor, a hydrogen separating membrane, and an atomizer/droplet ejection array.
Task Description:	The key objectives include (1) design, fabrication, and testing of the autothermal MEMS catalytic reactor integrated with hydrogen separating membrane, (2) design, fabrication, and experimental characterization of the novel micromachined fuel ejector array for energy-efficient atomization and mixing of liquid fuels, and (3) integration and experimental characterization of an integrated microreformer that combines a microreactor, a hydrogen separating membrane, and an atomizer/droplet ejection array. Space exploration is an energy intensive enterprise that demands power from Watt-scale for sensors and electronics to MegaWatt-scale for spaceship service and propulsion. Development of highly efficient and environmentally friendly power sources is equally important for the Earth-based applications. In addition, volume and mass limitations on spacecraft make the power density requirements very stringent (approaching 1.0-1.5 kWh per kilogram of fuel), especially for portable electronic and communication devices that are being used by astronauts during the extra vehicular activities. These requirements put an ever-increasing demand on development of small-scale, highly efficient “palm” power sources. The existing battery technologies become major obstacle to advances in performance of portable energy-intensive devices (primarily because of the limited lifetime between the recharges), and the fuel cell-based energy sources appear to be the only viable candidate for battery replacement. Fuel cell-based portable power sources offer opportunities for high efficiency and ultra-high density distributed power generation, which is essential for aerospace and terrestrial applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Pollution reduction and development of environmentally friendly energy conversion technology are the key benefits resulting from this research. We have developed novel technologies for liquid fuel atomization, hydrogen separation, and fuel reformation for hydrogen generation.
Task Progress:	We developed and characterized novel MEMS-enabled technologies for fuel processing for portable power generation and demonstrated the unique advantages offered by intimately integrating a microreactor with a micromachined ultrasonic atomizer for efficient dispersion of liquid fuels, and a micromachined Pd/Ag membrane for highly selective hydrogen separation. Periodic flow reversal at the appropriate frequency has been shown to be more robust to changes in operating conditions compared to a steady-state unidirectional flow. A valve-less flow reversal scheme integrated with a microchannel reactor has been developed to realize these advantages. An alternative design to the microchannel reactor, in which the atomized fuel impinges directly on a catalyst layer has also been invented, constructed and tested, yielding promising results for methanol steam reforming and partial oxidation in a more compact package. Both designs depend on the efficient atomization of liquid fuel supplied by the ultrasonic atomizer. The design, fabrication, and testing of this atomizer has been performed, demonstrating how precise control over the geometry results in a monodisperse stream of micrometer size fuel droplets when driven by a piezoelectric transducer on an acoustic resonance of the fluid cavity. Incorporating the hydrogen separation membrane in the reactor delivers pure hydrogen directly to the fuel cell anode and shifts the reaction equilibrium to increased hydrogen production. Process for fabrication of the nanostructured Pd/Ag metal alloy membrane has been developed, along with extensive test results showing excellent hydrogen permeation rates as well as remarkable long term stability.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Abstracts for Journals and Proceedings	McLeod L, Degertekin FL, Fedorov A. "Transient behavior of hydrogen permeation through Pd/Ag alloy micro/nano membranes." Presented at International Mechanical Engineering Congress & Exposition IMECE'06, Chicago, IL, November 5-10, 2006. International Mechanical Engineering Congress & Exposition IMECE 2006. , Nov-2006
Abstracts for Journals and Proceedings	Meacham JM, Varady MJ, Degertekin FL, Fedorov AG. "Fuel atomization from a micromachined ultrasonic droplet generator: visualization and scaling." Presented at International Mechanical Engineering Congress & Exposition IMECE'06, Chicago, IL, November 5-10, 2006. International Mechanical Engineering Congress & Exposition IMECE 2006. , Nov-2006
Abstracts for Journals and Proceedings	Varady MJ, McLeod L, Meacham JM, Degertekin FL, Fedorov AG. "MEMS-enabled processing of liquid fuels for fuel cell applications." Presented at 9th International Conference on Microreaction Technology (IMRET 9), Potsdam/Berlin, Germany, September 6-8, 2006. International Conference on Microreaction Technology IMRET 9 Proceedings, September 2006. , Sep-2006
Articles in Peer-reviewed Journals	Kaisare NS, Lee JH, Fedorov AG. "Operability analysis and design of a reverse-flow microreactor for hydrogen generation via methane partial oxidation." Industrial & Engineering Chemistry Research 2005 Oct;44(22):8323-33. http://dx.doi.org/10.1021/ie050231k , Oct-2005

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Articles in Peer-reviewed Journals	McLeod L, Degertekin FL, Fedorov A. "Non-ideal absorption effects on hydrogen permeation through palladium–silver alloy membranes." Journal of Membrane Science, 2009 Sep;339(1-2):109-14. http://dx.doi.org/10.1016/j.memsci.2009.04.036 , Sep-2009
Awards	Fedorov AG. "ASME Bergles-Rohsenow Young Investigator Award for Sustained Contribution to Heat, Mass, and Radiation Transfer (2007), November 2007." Nov-2007
Dissertations and Theses	Meacham JM. "A micromachined ultrasonic droplet generator: design, fabrication, visualization, and modeling." PhD Dissertation, Georgia Institute of Technology, August 2006. , Aug-2006
Papers from Meeting Proceedings	Varady MJ, McLeod L, Meacham JM, Degertekin FL, Fedorov AG. "Integrated MEMS infrastructure for fuel processing, hydrogen generation and separation for portable power generation." PowerMEMS 2006 Workshop, Berkeley, CA, November 29-December 1, 2006. Proceedings, PowerMEMS 2006 Workshop, December 2006. , Dec-2006
Patents	US Patent Application. Pending, November 2007. Nov-2007 Fedorov AG, Varady M, Degertekin FL. "Droplet Impingement Chemical Reactors and Methods of Processing Fuel."